



**Verified Carbon  
Standard**

## NKHOTAKOTA SOLAR PV PROJECT



Document Prepared by South Pole Carbon Asset Management Ltd

(South Pole)

|               |                                                                                                                                                                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Title | Nkhotakota Solar PV project                                                                                                                                                                                                                                          |
| Version       | 01                                                                                                                                                                                                                                                                   |
| Date of Issue | 26 May 2023                                                                                                                                                                                                                                                          |
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# 1 PROJECT DETAILS

## 1.1 Summary Description of the Project

The proposed project activity is a greenfield, grid connected solar PV based electricity generation project, located in the Least Developed Country (UNCTAD<sup>1</sup>) of Malawi, Africa. The purpose of this project is to provide clean energy to the local power grid as per design specifications and bolster renewable energy penetration in the relatively remote areas of Malawi, thereby improving energy access. This project is a voluntary project undertaken by the project developer Serengeti Nkhokota Ltd. Further this project is not required or mandated the local laws of the Government of Malawi. It is configured as a single instance of project activity and is not a grouped project.

With an electrification rate of only around 11% (UNCTAD<sup>2</sup>), Malawi has one of the most severely constrained power sectors in sub-Saharan Africa. Total system capacity stands at just 532<sup>3</sup> MW.

This Project entails the development of a 21 MWac / 26 MWp grid-connected solar PV plant, associated infrastructure including the sub-station and overhead power lines to connect to the Malawian ESCOM's 66 kV transmission line with the aim to bolster the Malawian grid's access to clean and renewable energy.

Pre project Scenario and baseline scenario are the same

The equivalent amount of electricity generated from the solar pv project would have been generated by the Malawian grid connected power plants encompassing fossil fuel-based power plants and stand by diesel power plants.

The project activity helps in reducing the GHG emission by replacing the equivalent quantity of fossil fuel fired grid connected power plants thereby, reduces 22,599 tCO<sub>2</sub>e per year on average with a total of 1,58,156 tCO<sub>2</sub>e over the first crediting period of 7 years.

## 1.2 Sectoral Scope and Project Type

The project activity falls under the following Sectoral scope and Project Type:

Sectoral Scope: 01 - Energy industries (renewable / non-renewables sources)

Project Type : I - Renewable Energy Projects

Methodology: ACM0002: Grid-connected electricity generation from renewable sources - Version 21<sup>4</sup>.

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<sup>1</sup> <https://unctad.org/topic/least-developed-countries/list>

<sup>2</sup> <https://unctad.org/topic/least-developed-countries/list>

<sup>3</sup> Malawi National Renewable Energy Policy

<sup>4</sup> ACM0002 v21

### 1.3 Project Eligibility

As per the latest applicable United Nations Datasets<sup>5</sup>, Malawi is a Least Developed Country, therefore a persuasive case is made that the project is eligible under the scope of the VCS Program Standard (V 4.4) with reference to Section 2.1.1, 2.1.2 and 2.1.3 namely:

- It results in CO2 emission reductions, one of the seven Kyoto Protocol greenhouse gases.
- It consists in grid-connected electricity generation using a greenfield solar power plant in a Least Developed Country
- The project activity follows the approved methodology's definition of project area and boundary. The current project is a single installation of greenfield solar PV of 21 MWac capacity.
- The project is a greenfield solar PV project in an LDC with very low energy access, therefore the project has been developed with the purpose of creating renewable energy for injection into the Malawian Grid. As Malawi is Least Developed Country as designated by the United Nations, the project does not belong to the projects excluded in Table 1 of VCS Standard.

Further, the project's eligibility and applicability under the ACM0002 Methodology is explained in detail under Section 3.2 of this Project Design Document.

### 1.4 Project Design:

- ☒ The project includes a single location or installation only.
- ☐ The project includes multiple locations or project activity instances but is not being developed as a grouped project
- ☐ The project is a grouped project.

#### Eligibility Criteria

This project has been designed as a single installation of project activity. It is not a group project.

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<sup>5</sup> <https://unctad.org/topic/least-developed-countries/list>

## 1.5 Project Proponent

|                       |                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------|
| <b>Organization</b>   | Serengeti Nkhotakota Limited                                                                   |
| <b>Contact person</b> | Wilfred Van Den Bos                                                                            |
| <b>Title</b>          | Development Director                                                                           |
| <b>Address</b>        | Realty House, Churchill Road, Limbe, Malawi                                                    |
| <b>Telephone</b>      | -                                                                                              |
| <b>Email</b>          | <a href="mailto:wilfredvandenbos@serengetienergy.com">wilfredvandenbos@serengetienergy.com</a> |

## 1.6 Other Entities Involved in the Project

|                            |                                                                            |
|----------------------------|----------------------------------------------------------------------------|
| <b>Organization name</b>   | South Pole Carbon Asset Management Ltd                                     |
| <b>Role in the project</b> | Carbon Consultant                                                          |
| <b>Contact person</b>      | Rohit Garg                                                                 |
| <b>Title</b>               | Regional Director, Climate Projects - South Asia and Middle East           |
| <b>Address</b>             | Technoparkstrasse 1, 8005 Zurich, Switzerland                              |
| <b>Telephone</b>           | +41-43 501 35 50                                                           |
| <b>Email</b>               | <a href="mailto:registration@southpole.com">registration@southpole.com</a> |

## 1.7 Ownership

Serengeti Nkhotakota Limited owns the project and will demonstrate ownership of the land and the project with the requisite documents issued to them. The ownership of project activity can be verified through following documents from the government-authorised bodies:

- Commissioning Certificate & Energy generation statement issued by State Electricity Board

## 1.8 Project Start Date

The project's expected date of commissioning is 10<sup>th</sup> June 2023 for commercial operations.

## 1.9 Project Crediting Period

7-year crediting period from date of project commissioning.

This project adopts twice renewable crediting period for a total of 21 years, and the First Crediting Period's Start Date is: 10/06/2023 (Expected date of Commissioning).

The planned lifespan of this project is expected to be 25 years.

## 1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO<sub>2</sub>e/year
- ☒ 20,000 – 100,000 tCO<sub>2</sub>e/year
- ☐ 100,001 – 1,000,000 tCO<sub>2</sub>e/year
- ☐ >1,000,000 tCO<sub>2</sub>e/year

| Project Scale |     |
|---------------|-----|
| Project       | (✓) |
| Large project |     |

| Year                                   | Estimated GHG emission reductions or removals (tCO <sub>2</sub> e) |
|----------------------------------------|--------------------------------------------------------------------|
| Year 1                                 | 23,471                                                             |
| Year 2                                 | 23,312                                                             |
| Year 3                                 | 23,103                                                             |
| Year 4                                 | 22,792                                                             |
| Year 5                                 | 22,385                                                             |
| Year 6                                 | 21,886                                                             |
| Year till 7 <sup>h</sup> year          | 21,301                                                             |
| <b>Total estimated ERs</b>             | 1,58,196                                                           |
| <b>Total number of crediting years</b> | 7                                                                  |
| <b>Average annual ERs</b>              | 22,599                                                             |

## 1.11 Description of the Project Activity

The project uses photovoltaic solar power technology (PV). Under the current project, the set up involves 71,220 Longi model LR6-72BP-370M solar PV panels to generate electricity by converting solar power into electric power to create clean energy with no consumption of fossil fuels and no emission of greenhouse gases. This is then converted from DC to AC using string inverters of Huawei SUN2000-105KTL-H1 make.

The solar PV panels in each array are connected in parallel and/or in string to get the required current and voltage which match with the rated input of the inverter. The inverter connected in each array converts the DC current to AC current. The electricity collected from all the inverters is stepped up to 66 kV by using Step Up Transformer. It then is exported to the national grid through a metering system.

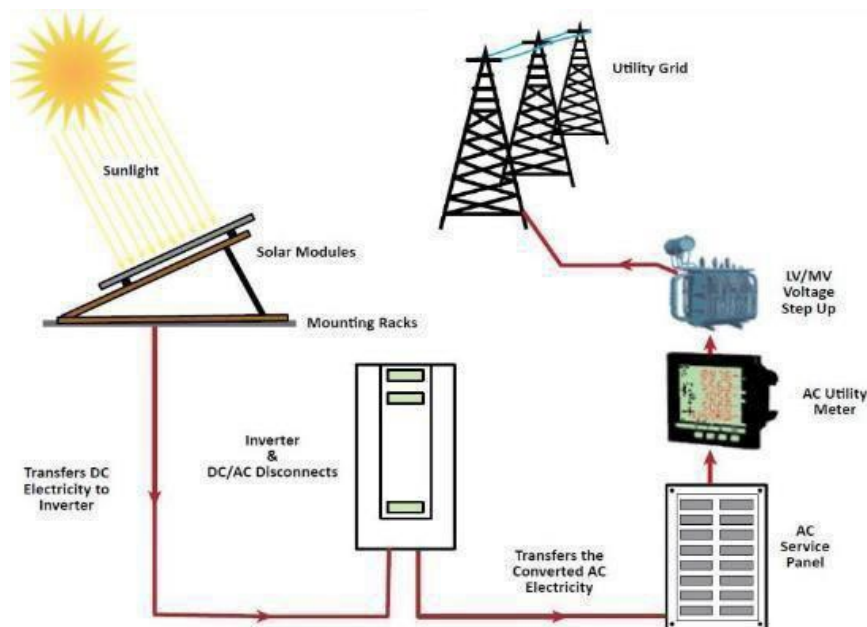


Image 1 : Simple Single Line Diagram for project's layout, source : detailed project report

**Table1:PV Plant Technology Specification and Configuration**

| Technical Parameter | Model type                          | Quantity | Specification                            |
|---------------------|-------------------------------------|----------|------------------------------------------|
| Solar PV Panels     | LONGi LR6-72BP-370M Bifacial module | 71,220   | 370 Watt per panel<br>(25-year lifespan) |
| Inverter            | Huawei SUN2000-105KTL-H1            | 198      | 105 KW max voltage output                |



**Table 2: Nkhotakota Solar PV farm Configuration**

| Parameter             | Values as required | Value as per design                                   |
|-----------------------|--------------------|-------------------------------------------------------|
| Plant export capacity | 21 MW              | 20.79 MW at 40°C                                      |
| Network Voltage (kV)  | 66 kV grid voltage | 11 kV stepped up to 66kV<br>(with 25 MVA transformer) |
| Operating frequency   | 50 Hz              | 50Hz                                                  |

### 1.12 Project Location

The site is located approximately 4km via road to the northwest of Nkhotakota, Malawi just off the M18 road between Nkhotakota and Kasungu. The site is approximately 60 hectares with overall dimensions of ~0.8km long by ~0.74km wide. With reference to below mentioned Image 2 and 3 the PV field is built within the rectangular section of the site.

The project site's coordinates are 12°57'50.4"S 34°15'19.9"E.



Image 2 : Project location, source : Google Maps image



Image 3 : Current project set up, source : project proponent

### 1.13 Conditions Prior to Project Initiation

The project is a green field solar PV installation, located in the Least Developed Country of Malawi (UNCTAD<sup>6</sup>). As per the US AID, the existing operating capacity of ESCOM of 532 MW, it is heavily dependent on hydro power, this is an issue as the country's dams frequently face drought and low water levels (USAID<sup>7</sup>). While most of the Malawi's power is hydro-based, efforts are underway to diversify generation sources with sources such as renewable solar PV and provide the local communities with energy.

A 'without project alternative' would place greater reliance on generating additional power from fossil fuel sources but also deny the local population potential employment opportunities associated with the Project. In this respect, the area would therefore continue along a marginalization path, with high poverty and unemployment levels, poor health facilities, poor access roads and poor state infrastructure.

### 1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Project Owner has received necessary approvals for development and commissioning for solar projects from the State agencies and follows the local laws and regulations.

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<sup>6</sup> <https://unctad.org/topic/least-developed-countries/list>

<sup>7</sup> <https://www.usaid.gov/powerafrica/malawi>

The project follows relevant permit regime under the Malawian Law. This includes an in-depth third part Environmental Social Impact Assessment and water abstraction permit. The project is a voluntary undertaking by the project proponent and is not mandated by any relevant local laws.

**Table 3 : Necessary Licenses Obtained by Project**

| License/Permit                         | Authority                                    | Date issued                    | Status                       |
|----------------------------------------|----------------------------------------------|--------------------------------|------------------------------|
| Environmental Impact Assessment        | Ministry of Environment                      | 19 <sup>th</sup> October 2018  | Permit issued with condition |
| Water Abstraction                      | Ministry of Irrigation and Water Development | 16 <sup>th</sup> May 2018      | Permit Issued                |
| General Waste Handling Permit          | Malawian Environment Protection Authority    | 6 <sup>th</sup> March 2023     | In Progress                  |
| Occupational Health and Safety License | Ministry of Labor                            | 14 <sup>th</sup> February 2023 | Issued                       |

## 1.15 Participation under Other GHG Programs

This project is not participating under any other GHG programs.

### 1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

This project is not seeking registration not been granted registration on any other GHG program.

### 1.15.2 Projects Rejected by Other GHG Programs

This project has not been registered and therefore not been rejected by any other GHG program.

## 1.16 Other Forms of Credit

### 1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

If yes, provide the name of the emissions trading program or other mechanism that allows GHG allowance trading.

### 1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

If yes, provide the name of the other program(s) under which the project has sought or received another form of GHG-related credit.

### 1.16.3 Supply Chain (Scope 3) Emissions

Have the owner(s) or retailer(s) of the impacted goods and services posted a public statement saying, "VCUs may be issued for the greenhouse gas emission reductions and removals associated with [organization name(s)] [name of good or service]" since the project's start date?

☐ Yes ☒ No

Since this is a grid connected greenfield Solar PV plant, hence scope 3 emissions within the supply chain are not valid under section 3.23.8 of the VCS V4.4, therefore there are no scope 3 emissions associated with this project.

Has the project proponent posted a public statement saying, "VCUs may be issued for the greenhouse gas emission reductions and removals associated with [name of good or service][describe the region or location, including organization name(s), where practicable]."

☐ Yes ☒ No

The project is awaiting commissioning; therefore, no such statement has been issued.

Have the producer(s) or retailer(s) of the impacted good or service been notified of the project and the potential risk of Scope 3 emissions double claiming via email?

☐ Yes ☒ No

As this is a grid connected greenfield Solar PV plant, hence scope 3 emissions within the supply chain are not valid under section 3.23.8 of the VCS V4.4, therefore there are no scope 3 emissions associated with this project.

### 1.17 Sustainable Development Contributions.

The project works on renewable energy in Malawi which is a Least Developed Country. Further the specifically selected site has very low levels of electrification, industry and formal employment. The Nkhotakota project will provide clean and renewable energy to the local populace and the national grid serviced by it, thereby reducing the use of diesel Peaker power plants while expanding the role of renewables. This will lead to better health, environmental and industrial outcomes as detailed below:



The project will install a greenfield Solar PV power plant in a remote region of Malawi and bolster the National Grid of Malawi (ESCOM<sup>8</sup>) with clean and renewable energy. The project will achieve a GHG reduction of 22,599 tCO<sub>2</sub>e/year during the first crediting period and therefore contribute to SDG 13 Climate Action



The construction and operation phase of this project will create new high quality jobs for which people will be drawn from the nearby communities, which will have a positive impact for the local economy. Therefore, the project will achieve SDG 8 Decent Work and Economic Growth. Currently 25 new jobs have been created for the local communities.



As per the current project design and layout the project will generate an average of 56 GWhrs annually<sup>9</sup> of clean, renewable energy over the first crediting period, providing the local communities with affordable energy access thereby achieving SDG 7 Affordable and Clean Energy.

## 1.18 Additional Information Relevant to the Project

### 1.18.1 Leakage Management

This section is not applicable as this is a greenfield grid connected solar PV project.

### 1.18.2 Commercially Sensitive Information

No commercial information currently under the scope of this PDD.

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<sup>8</sup> <http://www.escom.mw/index.php#>

<sup>9</sup> Detailed Project Report and PPA

### 1.18.3 Further Information

No further information is applicable under the current PDD.

## 2 SAFEGUARDS

### 2.1 No Net Harm

There is no net harm identified under this project, therefore no mitigation measures apply. The Project Proponent (PP) had an in depth ESIA conducted prior to the commencement of this project. Since this is a solar PV project, there are no adverse emissions or leakages or contamination of the air, water or local soil under this project. Furthermore, the project proponent will be recruiting the operations and maintenance staff from the nearby areas thereby generating high quality local employment.

### 2.2 Local Stakeholder Consultation

The project proponents conducted Local Stakeholder Consultations before the commencement of the project in line with the requirements of Verra Guidance. Initial consultations were carried out at a meeting held in Nkhotakota on 17th January 2018, between representatives of the local District Commissioner (DC), MITC representatives, the local Traditional Authority (TA), Senior Group Village Head (SGVH) and the local Village Heads. The meeting was also attended by the Project Proponent team. The meeting served to provide stakeholders with information about the Project and hear the local communities' perspectives for the project.

Stakeholders that were consulted during the Scoping phase of the project included the following:

- Department of Environmental Affairs (DEA)
- Malawian Investment Trade Centre (MITC)
- Nkhotakota District Commissioner's (DC) Office
- Representative from Nkhotakota District Lands Office
- Local Traditional Authority (TA)
- Senior Group Village Head (SGVH) (Head of the Village Heads)
- Heads of Local Villages

As part of this Local Stakeholder Consultation the following multi-method consultations was included and recorded:

- Public meetings—These involved consultation and engagement with residents from local community members, village chiefs and other interest groups.



- Focus group discussions—These involved consultation and engagement with village chiefs, a women’s group, an HIV-support group, a men’s group, a youth group and a business forum.
- Field observations- The physically inspected major facilities and landmarks, such as water supply points, school buildings, health care facilities, markets, occupational activities, informal sector activities and conditions of roads and houses. Other areas of interest studied were cultural heritage features of the project affected/neighboring communities. The demographic pattern, land size/use, economic activities (with emphasis on low-income groups highly dependent on primary activities), community structure, employment, markets, labor supply, income distribution, consumption, and migration pattern of the host/affected and neighboring communities was also discussed.



Image 4 and 5 The Local Stakeholder Consultation at project site, source: project documents

A continuous grievance mechanism has been established for the project as detailed in the ESIA report. The Continuous Grievance Mechanism is accessible to both employees and local community stakeholders. The project affected people are aware of the method to register their issues via the continuous grievance mechanism implemented and freely available at the project site. This is further monitored by the onsite HSE Team for satisfactory resolution of any issues registered.

Overall, following the above consultation activities the project was generally acceptable, with broad community support for the Project largely due to the recognized positive impacts to the socio-economic environment in the district and the wider country. Increased power generation from a renewable source that has very low negative impacts is seen to be of great benefit to the local community. The ESIA and LSC documents will be submitted to the DOE<sup>10</sup>.

## 2.3 Environmental Impact

The project activity instance under this project activity is using renewable energy generation technology (Solar energy) which is free from any kind of anthropogenic emission. Project activity is not having any negative environmental impacts.

<sup>10</sup> ESIA and LSC will be attached as reference document

A dedicated Environmental and Social Management plan has been developed and implemented by the project proponent team to mitigate any of the identified environmental impact from the project.

## 2.4 Public Comments

Under ESIA's comments, a grievance mechanism was agreed to be implemented under the Environmental and Social Management Plan<sup>11</sup>. This will ensure that all local stakeholders facing any issues can register their issue for resolution at any point of time throughout the lifetime of the project. Furthermore, the project will get listed at VERRA website for public comments as per the process and PP will consider all the comments appropriately.

# 3 APPLICATION OF METHODOLOGY

## 3.1 Title and Reference of Methodology

This project applies the VCS approved CDM methodology: ACM0002 Grid-connected electricity generation from renewable sources (Version 21.0). This methodology also refers to the latest approved version of the following tools:

- "Tool for the demonstration and assessment of additionality" (version 07.0.0)
- "Investment analysis" (version 11.0)
- "Common Practice Analysis" (version 3.1)

## 3.2 Applicability of Methodology

The project activity is a solar power project with a total installed capacity of 21 MWac hence "Consolidated baseline methodology for grid connected electricity generation from renewable sources" ACM0002 (Version 21) is applicable for this project. The project activity is installation of a solar PV power plant at project site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield Power Plant). The applicability of the methodology is as follows below:

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<sup>11</sup> Mentioned under ESIA and Livelihood Restoration Programme documents, which will be submitted.



**Table 4 : Applicability of ACM0002 methodology for project**

| Clauses | Requirements of the ACM0002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Scenario of the project                                                                                                 | Conclusion                        |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1       | <p>This methodology is applicable to grid- connected renewable energy power generation project activities that:</p> <ul style="list-style-type: none"> <li>a) Install a Greenfield power plant;</li> <li>b) Involve a capacity addition to (an) existing plant(s);</li> <li>c) Involve a retrofit of (an) existing operating plants/units;</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | The project is a greenfield grid-connected renewable power generation project.                                          | Applicability criterion satisfied |
| 2       | <p>The methodology is applicable under the following conditions:</p> <p>(a) The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit;</p> <p>In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.</p> | The project is a newly built solar power project                                                                        | Not Applicable                    |
| 3       | <p>In case of hydro power plants, one of the following conditions must apply:</p> <ul style="list-style-type: none"> <li>i) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or</li> <li>ii) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density calculated using equation (7), is greater than 4 W/m<sup>2</sup>; or</li> </ul> <p>The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m<sup>2</sup>; or</p>                                                                                                                                                                                                                                                                         | Not applicable, the project is not a hydro power plant, so this applicability condition does not need to be considered. | NA                                |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                |    |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 4 | <p>(d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m<sup>2</sup>, all of the following conditions shall apply:</p> <p>(i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m<sup>2</sup>;</p> <p>(ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity;</p> <p>(iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m<sup>2</sup> shall be:</p> <p>a. Lower than or equal to 15 MW; and<br/>Less than 10 per cent of the total installed capacity of integrated hydro power project.</p> | Not applicable, the project is not a hydro power plant, so this applicability condition does not need to be considered.                                                        | NA |
| 5 | <p>The methodology is not applicable to:</p> <p>a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; Biomass fired power plants/units.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | The project does not involve switching from fossil-fuels to renewable energy sources at the site of the project activity and the project is not a biomass-fired power project. | NA |
| 6 | <p>In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.</p>                                                                                                                                                                                                                                                                                                                                                                                               | Not applicable, the project is a newly built solar power project.                                                                                                              | NA |

### 3.3 Project Boundary

According to ACM0002 (Version 21.0), the spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the ESCOM (Malawian National Grid Operator) that the project power plant is connected to.

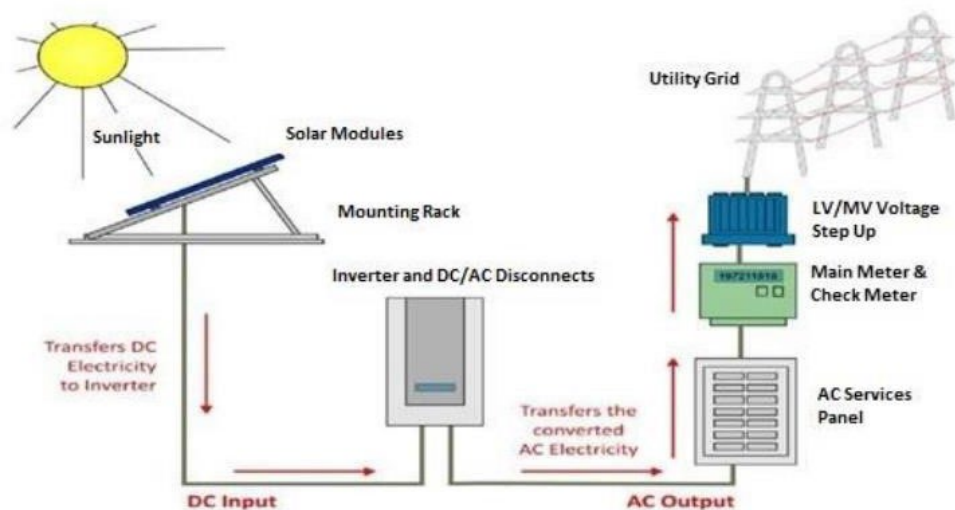


Image5 Simplified line diagram, source: project Detailed Project Report

Table 5 : Baseline Emissions Sources

| Source   |                                                     | Gas              | Included? | Justification/Explanation                                                                                                            |
|----------|-----------------------------------------------------|------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------|
| Baseline | Source 1<br>Emissions from fossil fuel power plants | CO <sub>2</sub>  | Yes       | Main emission source from fossil fuel power plants                                                                                   |
|          |                                                     | CH <sub>4</sub>  | Excluded  | Excluded for simplification. This emission source is assumed to be very small.                                                       |
|          |                                                     | N <sub>2</sub> O | Excluded  | Excluded for simplification. This emission source is assumed to be very small.                                                       |
|          |                                                     | Other            | Excluded  | Excluded for simplification. This emission source is assumed to be very small.                                                       |
|          | Source 2                                            | CO <sub>2</sub>  | Excluded  |                                                                                                                                      |
|          |                                                     | CH <sub>4</sub>  | Excluded  |                                                                                                                                      |
|          |                                                     | N <sub>2</sub> O | Excluded  |                                                                                                                                      |
|          |                                                     | Other            | Excluded  |                                                                                                                                      |
| Project  | Source 1<br>Solar PV                                | CO <sub>2</sub>  | Excluded  | No emissions from project                                                                                                            |
|          |                                                     | CH <sub>4</sub>  | Excluded  | As per the ACM0002, the production and operation of solar photovoltaic projects do not produce significant greenhouse gas emissions. |
|          |                                                     | N <sub>2</sub> O | Excluded  |                                                                                                                                      |
|          |                                                     | Other            | Excluded  |                                                                                                                                      |

| Source |          | Gas              | Included? | Justification/Explanation                          |
|--------|----------|------------------|-----------|----------------------------------------------------|
|        | Source 2 | CO <sub>2</sub>  | Excluded  | Thus, the project emissions can be safely ignored. |
|        |          | CH <sub>4</sub>  | Excluded  |                                                    |
|        |          | N <sub>2</sub> O | Excluded  |                                                    |
|        |          | Other            | Excluded  |                                                    |

### 3.4 Baseline Scenario

According to the ACM0002 Large Scale Consolidated Methodology Version 21: “If the project activity is the installation of a Greenfield power plant. The baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as mentioned in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system.”

In the absence of the project activity, the equivalent amount of power would have been drawn from the Malawian grid or generated from fossil fuels. Hence, the baseline for the project activity is the equivalent amount of power from the national grid.

Project activity is the generation of electricity from renewable energy sources. The power generated from the solar PV plant has zero emissions; there is no material leakage and the electricity generated will be fed into the fossil intensive national grid through the interconnection facility at the site.

### 3.5 Additionality

According to the methodology ACM0002 (Version 21.0), the additionality of the project activity shall be demonstrated and assessed using the latest version of the “TOOL01: Tool for the demonstration and assessment of additionality”. All calculations made shall be provided as annexures

The steps are listed as following:

**Step 0:** First of Its Kind project activities, this does not apply as there are other solar projects within the project country.

**Step 1:** Identification of alternatives to the project activity consistent with current laws and regulations. Define realistic and credible alternatives to the project activity(s) through the following Sub-steps.

**Sub-step 1a:** Define alternatives to the project activity: Alternatives available to the project participant or similar project developers that provide outputs or services comparable with the project activity are identified below:

(a) The proposed project activity undertaken without being registered as a VCS project activity.  
Alternative

(b) Continuation of the current situation (no project activity or other alternatives undertaken).

**Sub-step 1b:** Consistency with mandatory laws and regulations

Both alternatives (a) and (b) are realistic alternatives to the project, which are consistent with mandatory laws and regulations of Malawi.

**Step 2:** Investment analysis Sub-step 2a: Determine appropriate analysis method:

According to ACM0002 (Version 21.0), if the project activity and the alternatives identified in Step 1 generate no financial or economic benefits other than carbon related income, then apply the simple cost analysis (Option I). Otherwise, use the investment comparison analysis (Option II) or the benchmark analysis (Option III). For the project, the alternative to the project activity is the supply of electricity from a grid which is not considered as an investment by project participant, so the benchmark analysis (Option III) is used.

**Sub-step 2b:** Option III. Apply Benchmark Analysis

According to “Investment analysis” (version 8.0, in force during 2018), “For Malawi, the rate for the year of 2018 October is 18.57% for Group 1 projects (renewable power projects).<sup>12</sup>

**Sub-step 2c:** Calculation and comparison of financial indicators:

The relevant financial analysis is an Internal Rate of Return calculation from the project’s costing data. The equity IRR is therefore calculated on the parameters that are applicable to the Malawian market data, this will be considering the specific characteristics of the project activity and not linked to the profitability for the project. Based on the parameters including the financial parameters, loan, working capital, applicable corporate taxes, the EIRR sheet was developed, the benchmark equity IRR for Malawi for the year 2018 was determined to be 27.41% and the project’s was estimated to be 3.11%.

The project activity cannot be considered as financially attractive as the equity IRR for the project activity is less than the Malawian Benchmark EIRR.

**Table 7 Estimated Project IRR and Benchmark Values**

| Investment<br>Analysis –<br>Outcome | Equity IRR | Benchmark (Equity IRR) |
|-------------------------------------|------------|------------------------|
|                                     | 3.11%      | 27.41%                 |

<sup>12</sup> <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v8.pdf>

### Sub-step 2d: Sensitivity Analysis

The following parameters were used to develop the sensitivity analysis for this project :

1. PLF
2. O&M Cost
3. Project Cost
4. Tariff

Based on the above parameters, the ultimate objective of the sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the scenario presented, in order to provide a cross-check on the suitability of the assumptions used in the development of the investment analysis, the results are below :

**Table 8 : Sensitivity Analysis**

| <b>Variation %</b> | <b>-10%</b>   | <b>Normal</b> | <b>10%</b>    | <b>Variation required to reach benchmark</b> |
|--------------------|---------------|---------------|---------------|----------------------------------------------|
| <b>PLF</b>         | <b>9.34%</b>  | <b>3.11%</b>  | <b>14.75%</b> | <b>16.71%</b>                                |
| <b>Opex</b>        | <b>12.56%</b> | <b>3.11%</b>  | <b>11.35%</b> | <b>-74.20%</b>                               |
| <b>Capex</b>       | <b>14.34%</b> | <b>3.11%</b>  | <b>10.10%</b> | <b>-17.75%</b>                               |
| <b>Tariff Rate</b> | <b>9.34%</b>  | <b>3.11%</b>  | <b>14.75%</b> | <b>16.71%</b>                                |

The results of sensitivity analysis show that even with a variation of +10% & -10% in project cost, O&M cost, PLF and Tariff Rate , the Equity IRR is significantly lower than the benchmark. This is evident from the results given above; the project remains additional even under the most favorable conditions.

### Step 4: Common practice analysis

As the project applies power generation based on renewable energy which is one of the measure(s) listed in the definitions section above, we proceed to Sub-step 4a.

**Sub-step 4a:** The proposed project activity(ies) applies measure(s) that are listed in the definitions section above Common practice analysis is carried out as per “Common practice” (version 03.1).

**Sub-Step 4a-1:** calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity The installed capacity of the project is 26 MWp( DPR<sup>13</sup>), so the applicable capacity range as +/-50% of the total design capacity is 13 ~ 39 MWp.

**Step 4a-2:** identify similar projects which fulfil all of the following conditions: Malawi is selected as applicable geographical area. 1 project is identified which started commercial operations before within the required timeframe.

**Sub-step 4a-3:** Within the projects identified in Sub-step 4a-2, identify those that are neither registered CDM/VCS/CCER project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number as  $N_{all}$ .

According to Sub-Step 4a-2, there is 1 solar power plant<sup>14</sup> with capacity from 13 MWp to 39 MWp operational in Malawi ,  $N_{all} = 1$ .

Sub-step 4-4: Within similar projects identified in Sub-step 4-3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number  $N_{diff}$ .

For the project,  $N_{diff} = 1$ .

**Sub-step 4-5:** Calculate factor  $F = 1 - N_{diff}/N_{all}$  representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology like the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

Since  $N_{all} = 1$ ,  $N_{diff} = 1$ , therefore,  $N_{all} - N_{diff} = 0$ , which is equal to 0.  $F = 1 - N_{diff}/N_{all} = 0$ .

According to common practice (version 03.1), the proposed project activity is a “common practice” within a sector in the applicable geographical area if the factor F is greater than 0.2.

$N_{diff}$  is 1. However, the  $N_{all} - N_{diff}$  is 0 therefore less than 3. Thus, the Project activity is not a “common practice” as it satisfies the criteria laid.

Outcome of Step 4: Based on the analysis above, the baseline scenario of the project is: The proposed project can be considered as additional<sup>15</sup>.

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<sup>13</sup> As mentioned in DPR ,21MWac is grid injection capacity, 26MWp is total design capacity.

<sup>14</sup> Golomoti Solar Power Plant

<sup>15</sup> IRR,CPA sheets will be provided as per the laid out methodologies and tools.

# 4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

## 4.1 Baseline Emissions

According to the latest version of the methodology, ACM0002 Large Scale Consolidated Methodology: Grid-connected Electricity Generation from Renewable Sources, Version 21.0, since the project activity is the installation of Greenfield Renewable Energy Power Plant, the baseline scenario is as follows: If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected fossil fuel power plants and by the addition of new generation sources.

$EF_{grid,CM,y}$  = Combined margin CO<sub>2</sub> emission factor for grid connected power generation in year y calculated using the latest version of “TOOL07: Tool to calculate the emission factor for an electricity system” (tCO<sub>2</sub> / MWh).

The value taken is 0.397 tCO<sub>2</sub>e/MWhr<sup>16</sup> sourced from the IFI Harmonized Emission Grid Factors dataset v.3.2 for the country of Malawi. In case of Greenfield power plant development, the calculation of the electricity system's combined margin (CM) shall be based on the following equation, according to the methodology:

$$EG_{PJ,y} = EG_{Facility,y}$$

Where:

$EG_{PJ,y}$  = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)  $EG_{facility}$ .

y = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr).

## 4.2 Project Emissions

The project activity involves solar PV based renewable power generation which has no emissions over its lifetime, hence  $PE_y = 0$ .

## 4.3 Leakage

As this current project is a greenfield Solar PV project, in line with ACM0002, no leakage is considered for this project.

<sup>16</sup> IFI Harmonised Grid Emission Factor Dataset V.3.2



#### 4.4 Net GHG Emission Reductions and Removals

The equations used for Net GHG Emission Reductions are calculated as follows:

$$ER_y = BE_y - LE_y - PE_y$$

Where  $ER_y$  = Emission reductions in year y (tCO<sub>2</sub>e)

$BE_y$  = Baseline emissions in year y (tCO<sub>2</sub>e)

$PE_y$  = Project emissions in year y (tCO<sub>2</sub>e)

$LE_y$  = Leakage emissions in year y (tCO<sub>2</sub>e) Since  $LE_y = 0$  and

$PE_y = 0$   $ER_y = BE_y$

Table 9 : Net GHG Removals

| Year                                                        | Estimated baseline emissions or removals (tCO <sub>2</sub> e) | Estimated project emissions or removals (tCO <sub>2</sub> e) | Estimated leakage emissions (tCO <sub>2</sub> e) | Estimated net GHG emission reductions or removals (tCO <sub>2</sub> e) |
|-------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------|
| 10 <sup>st</sup> June 2023 - 09 <sup>th</sup> June 2024(Y1) | 23,417                                                        | 0                                                            | 0                                                | 23,417                                                                 |
| 10 <sup>st</sup> June 2024- 09 <sup>th</sup> June 2025(Y2)  | 23,312                                                        | 0                                                            | 0                                                | 23,312                                                                 |
| 10 <sup>st</sup> June 2025 – 09 <sup>th</sup> June 2026(Y3) | 23,103                                                        | 0                                                            | 0                                                | 23,103                                                                 |
| 10 <sup>st</sup> June 2026- 09 <sup>th</sup> June 2027 (Y4) | 22,792                                                        | 0                                                            | 0                                                | 22,792                                                                 |
| 10 <sup>st</sup> June 2027- 09 <sup>th</sup> June 2028(Y5)  | 22,385                                                        | 0                                                            | 0                                                | 22,385                                                                 |
| 10 <sup>st</sup> June 2028- 09 <sup>th</sup> June 2029(Y6)  | 21,886                                                        | 0                                                            | 0                                                | 21,886                                                                 |
| 10 <sup>st</sup> June 2029- 09 <sup>th</sup> June 2030(Y7)  | 21,301                                                        | 0                                                            | 0                                                | 21,301                                                                 |
| <b>Total</b>                                                | 1,58,196                                                      | 0                                                            | 0                                                | 1,58,196                                                               |

# 5 MONITORING

## 5.1 Data and Parameters Available at Validation

|                                                                                                     |                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data / Parameter</b>                                                                             | <b>EF<sub>grid,CM, y</sub></b>                                                                                                               |
| <b>Data unit</b>                                                                                    | tCO <sub>2</sub> e/MWhr                                                                                                                      |
| <b>Description</b>                                                                                  | Emission factor for Malawian Grid taken from UN dataset                                                                                      |
| <b>Source of data</b>                                                                               | UNFCCC IFI Harmonized Grid Emission Factors V3.2 Standardized Baseline                                                                       |
| <b>Value applied</b>                                                                                | 0.397 tCO <sub>2</sub> e/MWhr                                                                                                                |
| <b>Justification of choice of data or description of measurement methods and procedures applied</b> | Combined Margin emissions from IFI Harmonized Grid Emission Factors version 3.2 dataset is the only relevant and complete dataset available. |
| <b>Purpose of Data</b>                                                                              | Calculation of baseline emissions                                                                                                            |
| <b>Comments</b>                                                                                     | Not applicable                                                                                                                               |

## 5.2 Data and Parameters Monitored

|                                                                        |                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data / Parameter</b>                                                | <b>EG<sub>PJ,y</sub></b>                                                                                                                                                                                                                                                            |
| <b>Data unit</b>                                                       | MWh/yr                                                                                                                                                                                                                                                                              |
| <b>Description</b>                                                     | Quantity of net electricity generation supplied by the project plant/unit to the grid in year y                                                                                                                                                                                     |
| <b>Source of data</b>                                                  | Direct measurement of electricity supplied to the grid using project meters installed                                                                                                                                                                                               |
| <b>Description of measurement methods and procedures to be applied</b> | <p>The Net electricity supplied to the grid by the project activity will be calculated based on the energy delivered to &amp; received from the grid monitored and measured by Main Meter and backup Meter</p> <p>Meter Make : CeWe Pro meter 100</p> <p>Meter Model : Advanced</p> |

|                                          |                                                                                                                                                                                                                                    |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frequency of monitoring/recording</b> | Continuous monitoring and monthly recording                                                                                                                                                                                        |
| <b>Value applied</b>                     | MWh                                                                                                                                                                                                                                |
| <b>Monitoring equipment</b>              | CeWe Pro Meter 100 advanced of 0.2 class                                                                                                                                                                                           |
| <b>QA/QC procedures to be applied</b>    | Calibration of all the meters will be undertaken at yearly intervals and faulty meters will be duly replaced immediately. Calibration Frequency: Once a year or as per the PPA. The data will be cross checked with sales receipts |
| <b>Purpose of data</b>                   | Calculation of project emissions                                                                                                                                                                                                   |
| <b>Calculation method</b>                | Visual confirmation from main meter and check meter                                                                                                                                                                                |
| <b>Comments</b>                          | Not applicable                                                                                                                                                                                                                     |

### 5.3 Monitoring Plan

**The monitoring plan** for this project is developed in accordance with methodological requirements, project activity instances will be renewable energy-based power generation projects. The project instance is a grid-connected solar power project being implemented in Nkhotakota, Malawi. The monitoring plan, which will be implemented by the project participant describes the monitoring organization, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving.

**Roles and responsibilities of Facilities, General Manager:** Overall functioning and maintenance of the project activity, collect and manage all data and its archiving and shall coordinate with the O&M operator and project proponent.

**Project Personnel:** hired and trained to maintain the project facilities. The project personnel may be drawn from the local population for maintenance and routine management.

**QA/QC for Meters:** Project meter will be maintained and checked by Project Proponent and check meter will be maintained and checked by the Malawian ESCOM. The meters are sealed and can only be opened in the presence of representatives from both the project proponent and ESCOM.

The project proponent shall check the reading from the main meter on the last day of the month to check net electrical output.

| <b>Meter type</b> | <b>Serial Number</b> | <b>Make and Model</b>  | <b>Accuracy class</b> | <b>Date of installation</b>  |
|-------------------|----------------------|------------------------|-----------------------|------------------------------|
| Main Meter        | WP028188             | CeWe Pro meter-Advance | 0.2                   | 19 <sup>th</sup> April, 2023 |
| Check Meter       | WP028189             | CeWe Pro meter-Advance | 0.2                   | 19 <sup>th</sup> April, 2023 |

The records will be cross-checked with the records of sold electricity i.e., with the invoice. The meters will be calibrated at least once a year by the Project Proponent as per the terms of the PPA and TCA.

**Data Measurement of Project:** The export and import energy will be measured continuously using Main & check meter installed at inter connection point of project( as mention above). Net export will be calculated from the export & Import electricity readings of Main meter installed at project interconnection point and shall be taken on monthly basis by authorized officers of Project Proponent in the presence of representative of Malawian Grid and Project proponent if required.

**Data collection:** Export & Import readings from the meters will be collected under the supervision of the authorized representatives of PP. The net electricity supplied to the grid would be calculated based on Main Meter readings. Data would be recorded and stored in photographic and paper format not less than 7 years post end of PPA.

**Emergency preparedness** the project activity will not result in any unidentified activity that can result in substantial emissions from the project activity. No need for emergency preparedness in data monitoring is visualized. In the unlikely event of failure of all Main meter installed at site substation backup meter reading shall be used and all the faulty meters are required to be repaired or replaced simultaneously, the export readings from Main & backup Meters installed at the inter-connection point at the project site will be used for monitoring of net electricity exported to the grid.

**Personnel training:** In order to ensure the proper functioning of the project activity and a accurate monitoring of emission reductions, the staff will be trained. The plant helpers will be trained in equipment operation, data recording, reports writing, operation and maintenance and emergency procedures in compliance with the monitoring plan.